Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWI")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```r
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40, 2000);
Distrib (Fm,           TruncNormal,            0.1,      0.2,    0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,         0.8,      0.1,    0.1, 2);
Distrib (kap,          Uniform,                0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40, 300);
Distrib (E_coc,        TruncNormal_cv,              30,      0.2, 1, 300);
Distrib (rOM_ClxHorse,  LogUniform,           0.00001,      0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (w,  TruncNormal_cv,           40, 10,      5, 100);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcaliSWI DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	187.13500	0.4298800	0.9066130	0.59672300	86.11470	5.7509400
2.5%	155.63900	0.1116030	0.7755192	0.57822200	59.08730	3.9504600
97.5%	203.38545	0.6312776	1.0107200	0.63409140	107.64300	7.1765745
Min	82.85490	0.0903452	0.6993020	0.32306700	47.41400	3.3907700
Max	218.62800	0.7963090	1.0983900	0.65789400	130.62900	36.0381000

```
Result_mean 180.40335 0.3609949   0.9022442 0.60529667  79.44977  5.2381299
Result_sd    11.96534 0.1235058   0.0630484 0.01479198  12.01652  0.8627442
            rOM_ClxHorse.1. f_Slope.1.     w.1. Sigma_W.1.
Result_mode    0.0010604100 0.24703500 65.78230 0.12479000
2.5%           0.0002920040 0.18669400 46.58972 0.10721520
97.5%          0.0030935400 0.33809400 98.62390 0.14063000
Min            0.0000357106 0.15553200 35.52550 0.09591060
Max            0.0059429400 1.45139000 99.98870 6.89406000
Result_mean    0.0013500248 0.24933866 73.91086 0.12354236
Result_sd      0.0007320051 0.03654344 14.43358 0.05785914

Number of observations, n =  277
Number of parameters, k =  9
Log-likelihood, LL =  -14.27171
BIC =  79.15958


Potential scale reduction factors:

                Point est. Upper C.I.
E_coc.1.              1.27       1.74
Ehp.1.                1.41       2.07
f_Slope.1.            1.05       1.16
Fm.1.                 1.36       1.98
kap.1.                1.11       1.35
pAm.1.                1.07       1.20
r_pAm_pM.1.           1.38       2.01
rOM_ClxHorse.1.       1.01       1.02
Sigma_W.1.            1.00       1.01
w.1.                  1.26       1.73

Multivariate psrf

1.36
```

```r
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)
```

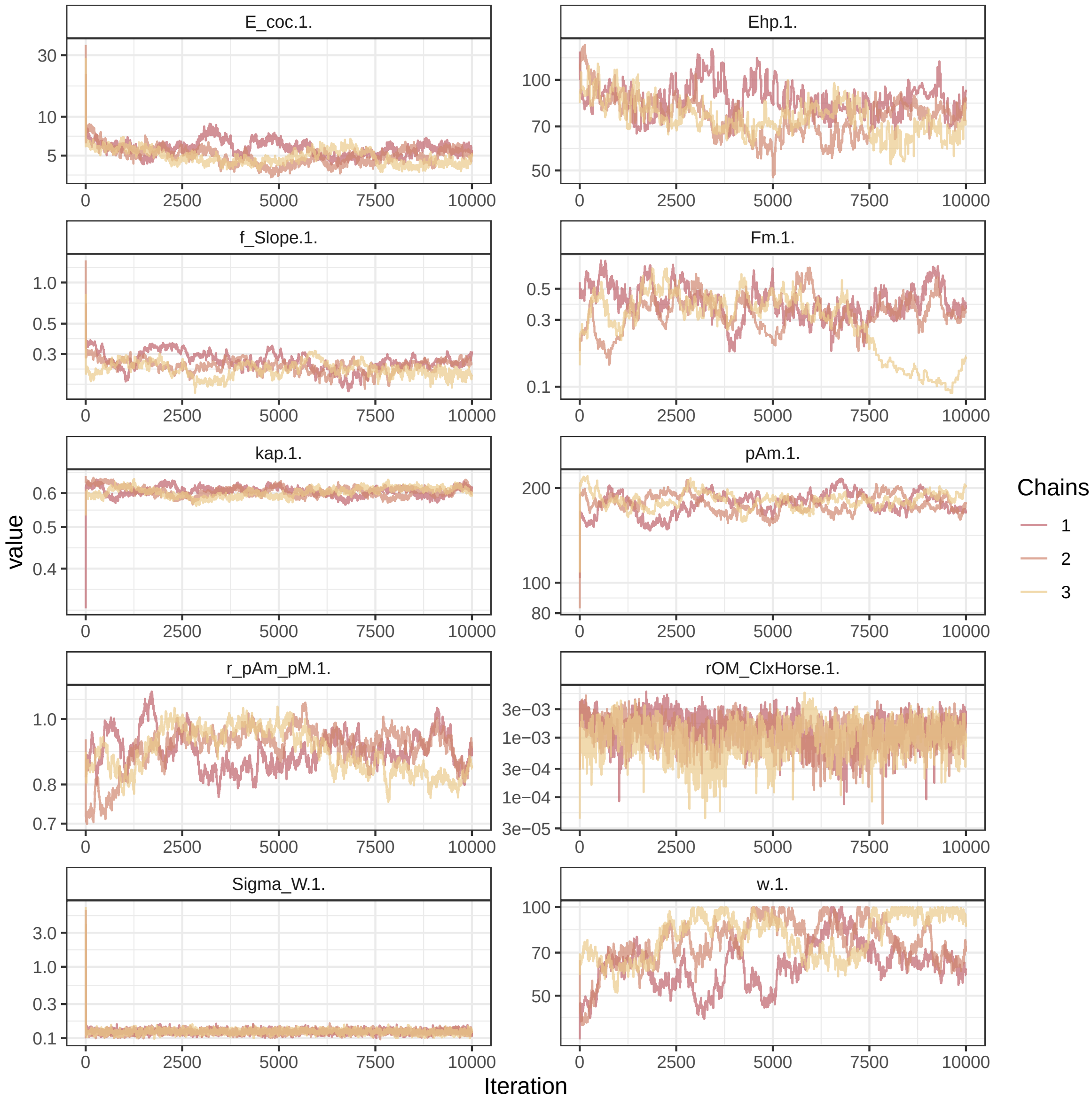


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

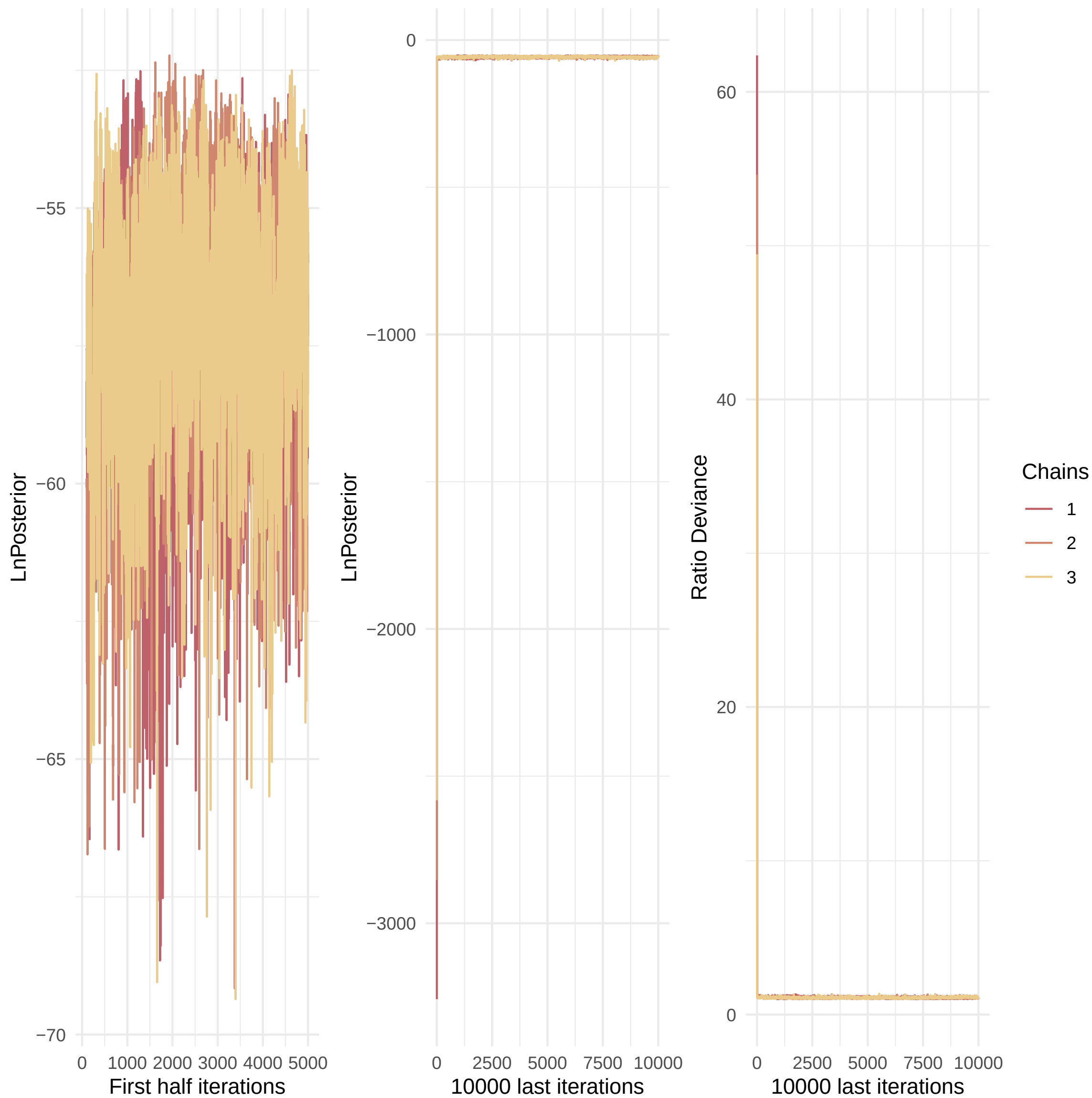


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

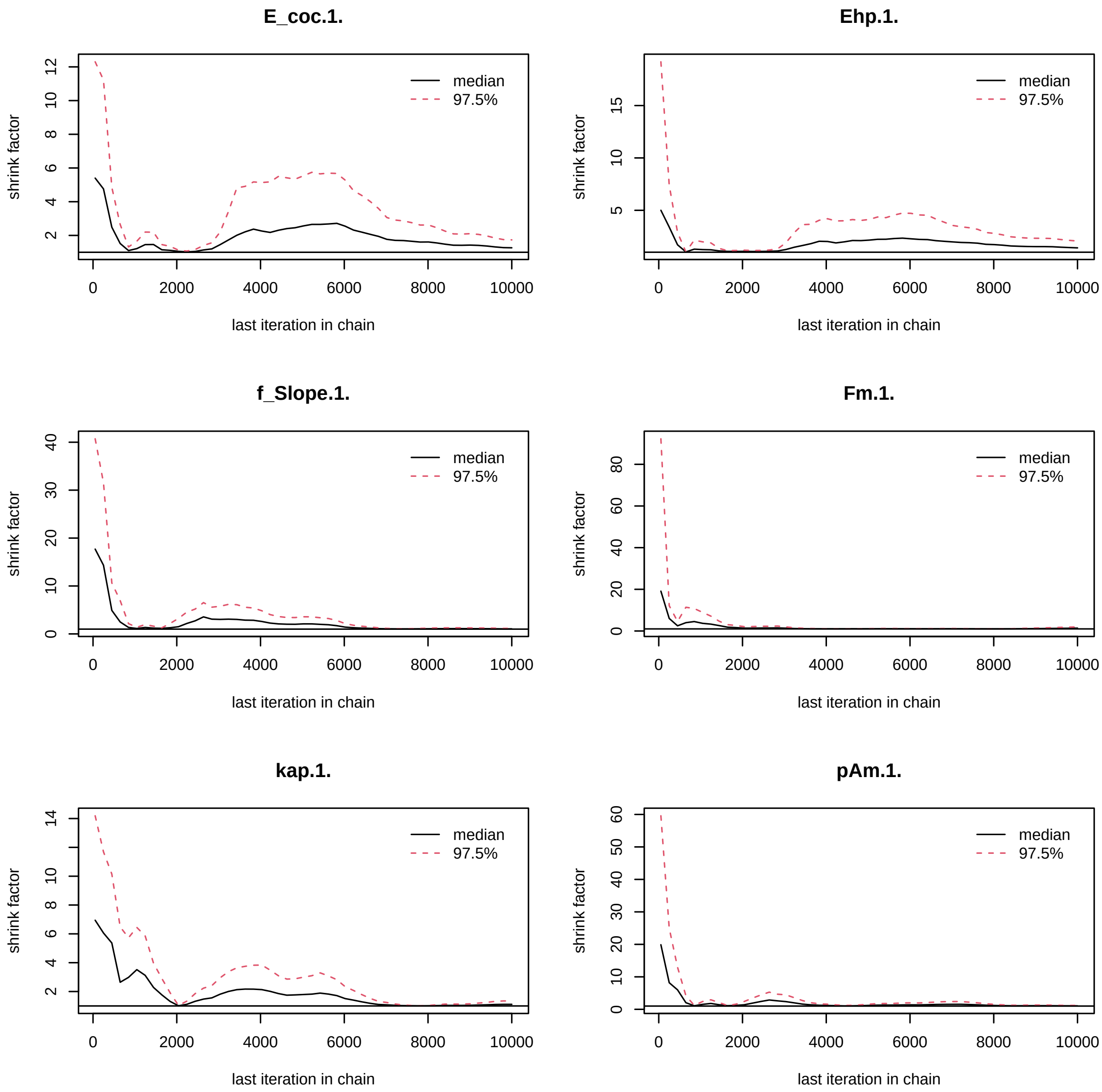


Figure 3: Chains convergence.

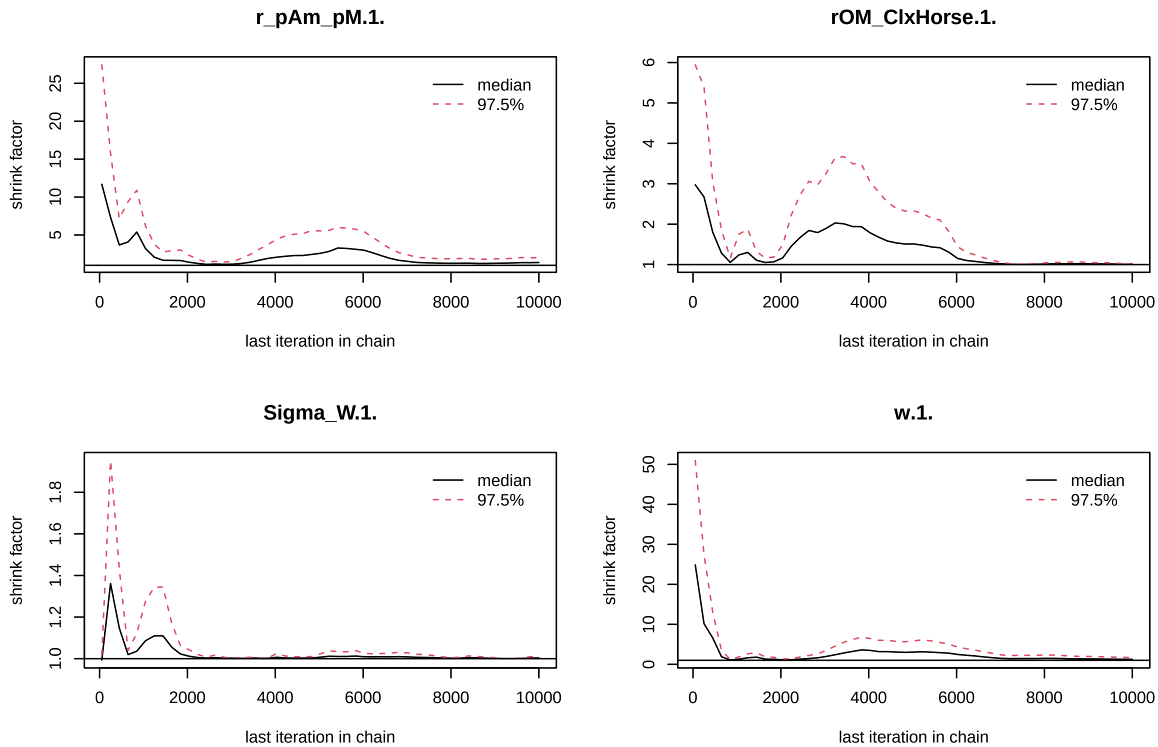


Figure 4: Chains convergence.

```
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

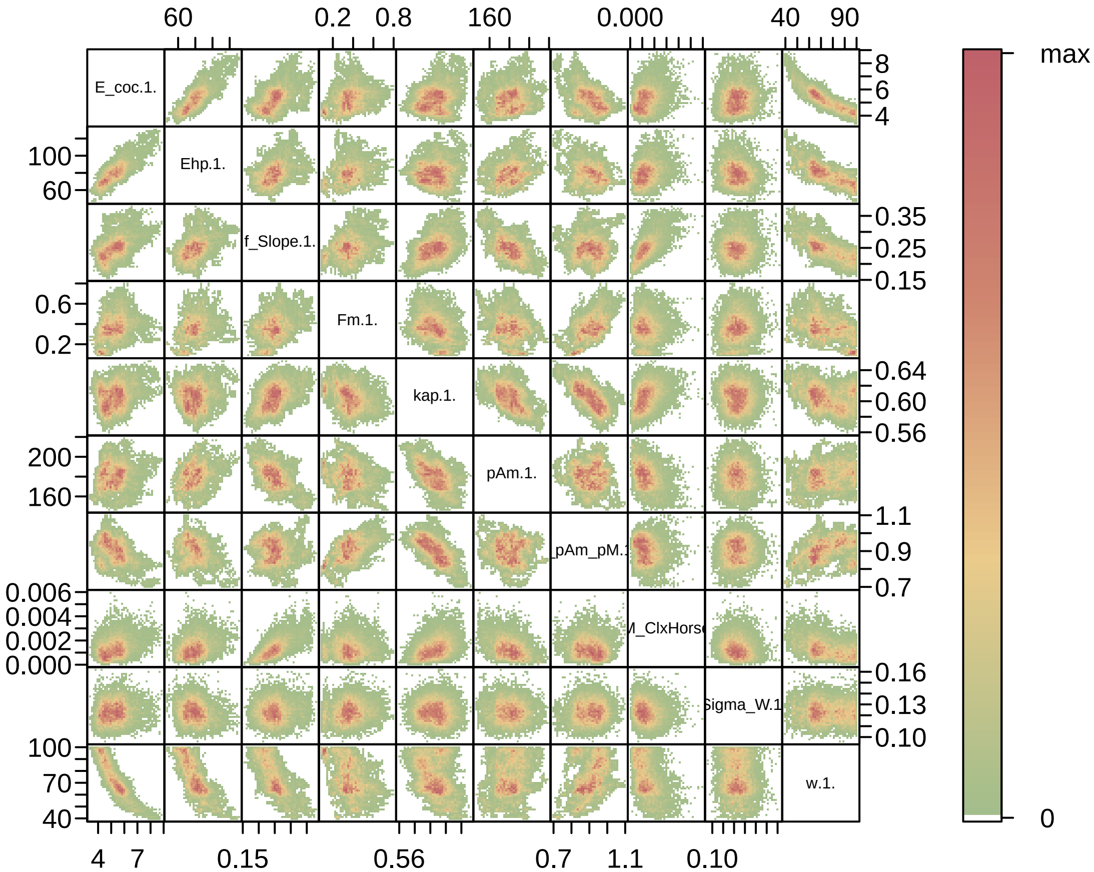


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
```

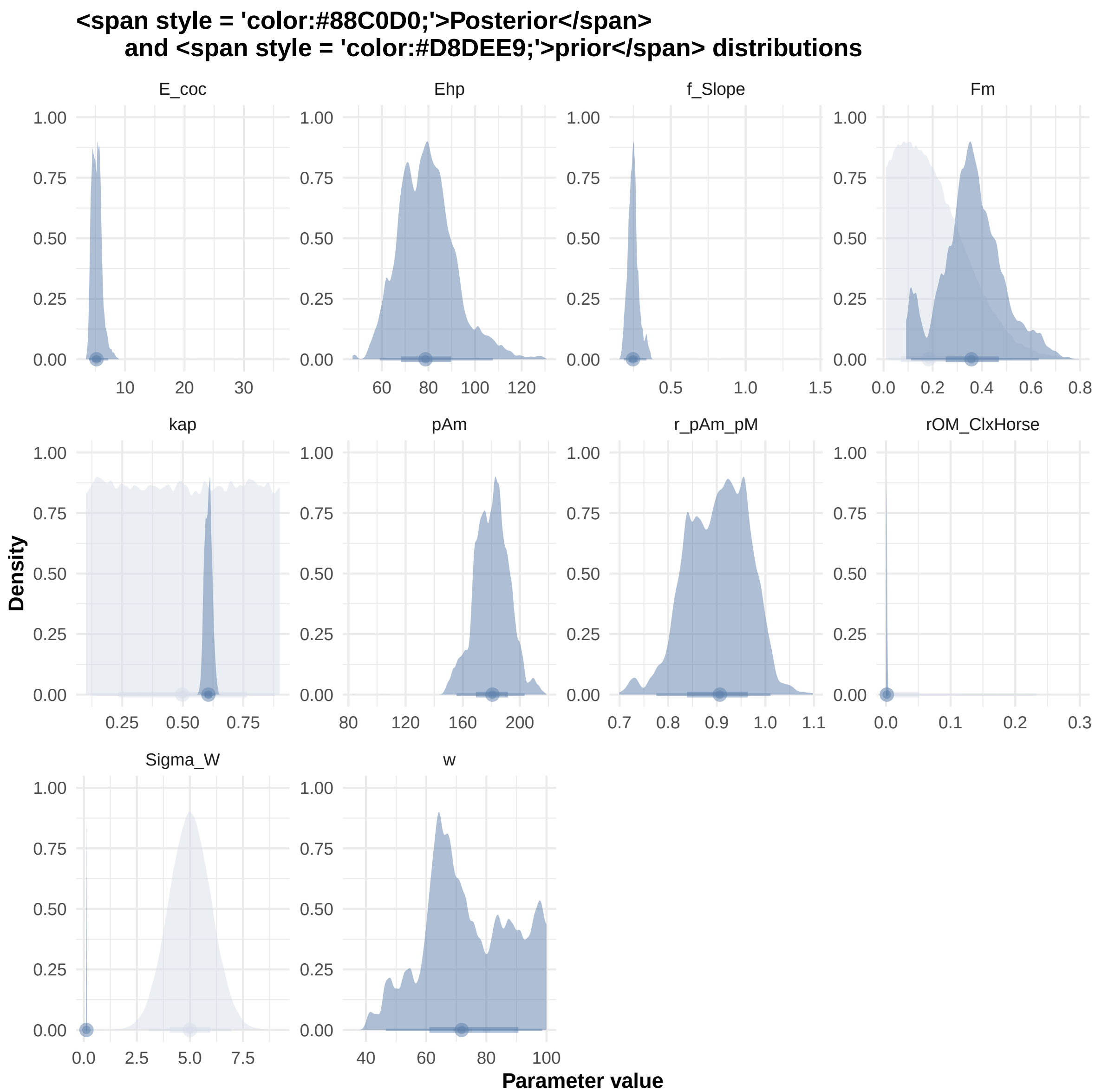


Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcaliSWI DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
```

```
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
##############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

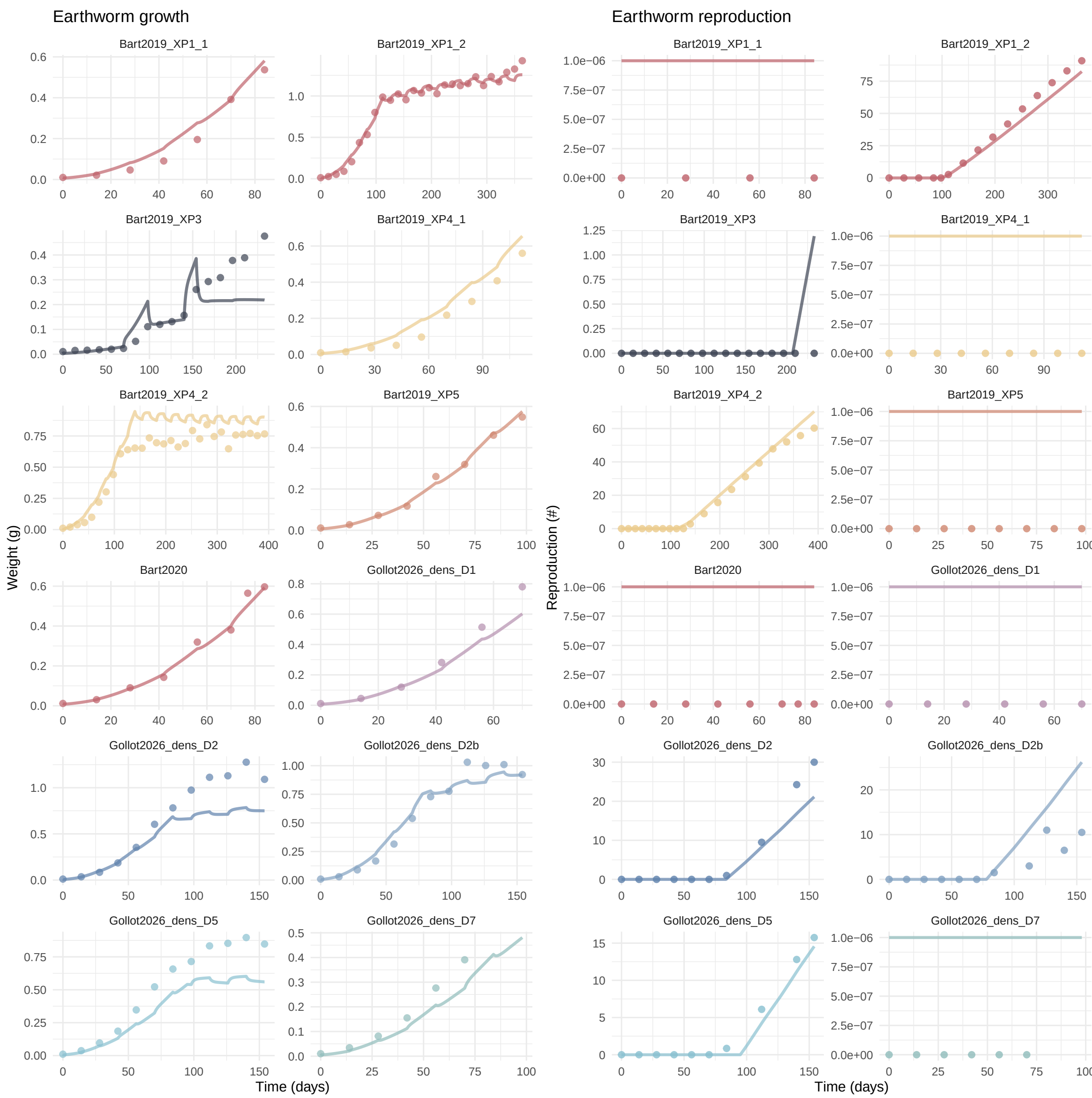
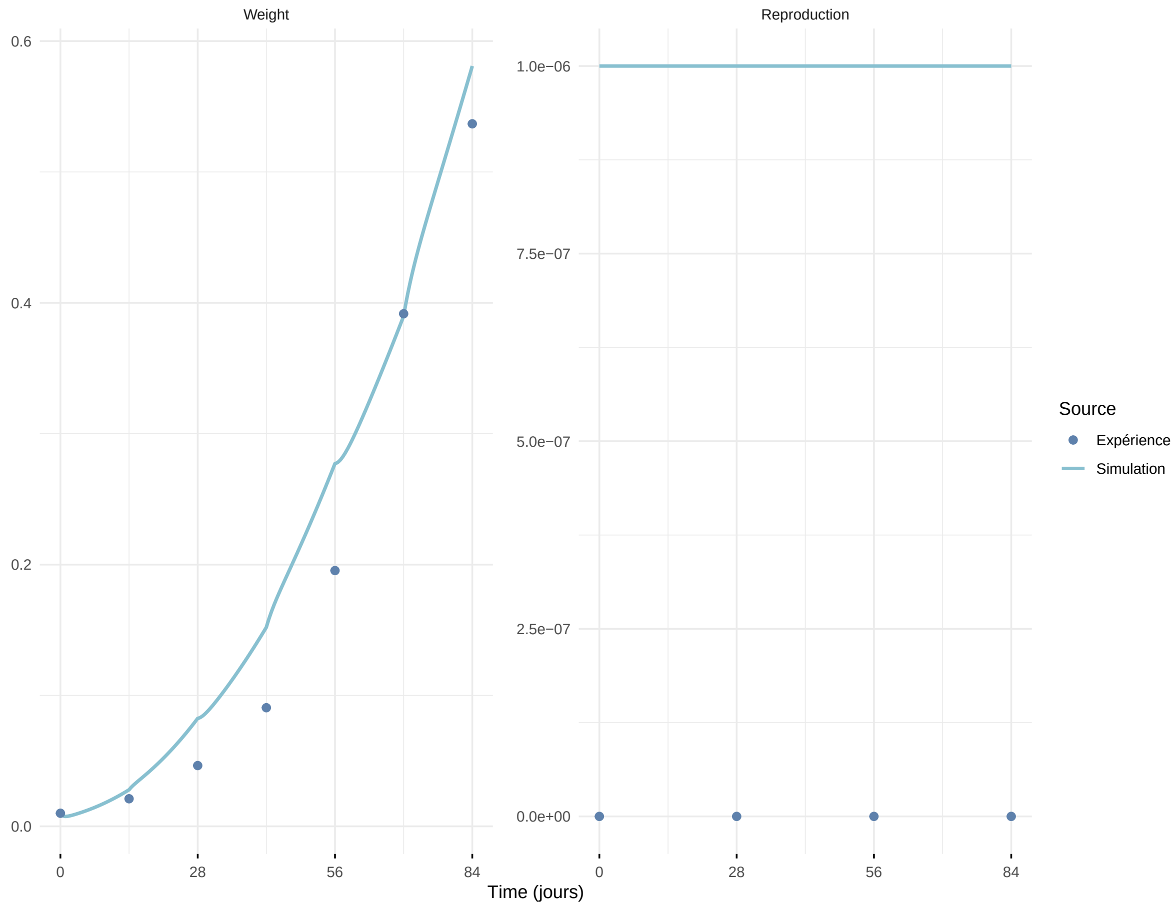
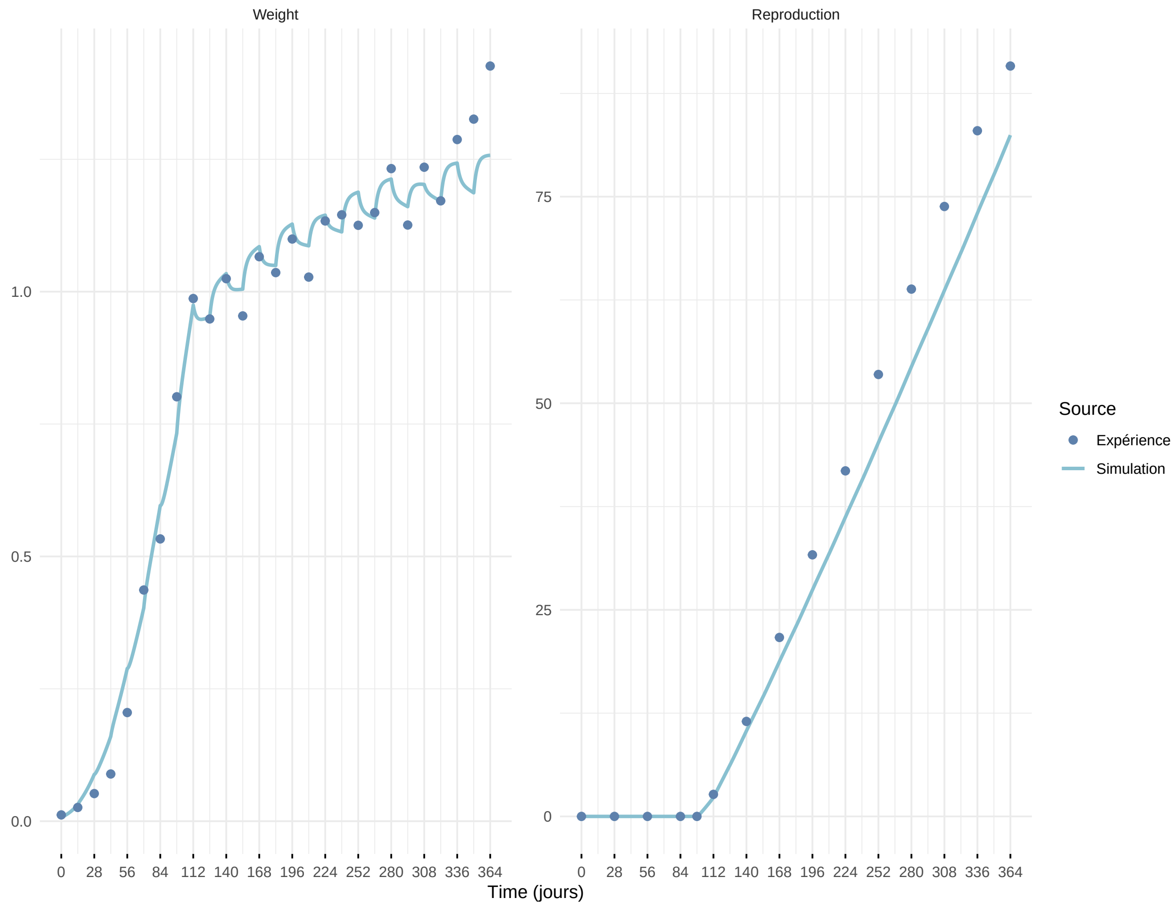


Figure 7: Simulations Weight and Reproduction.

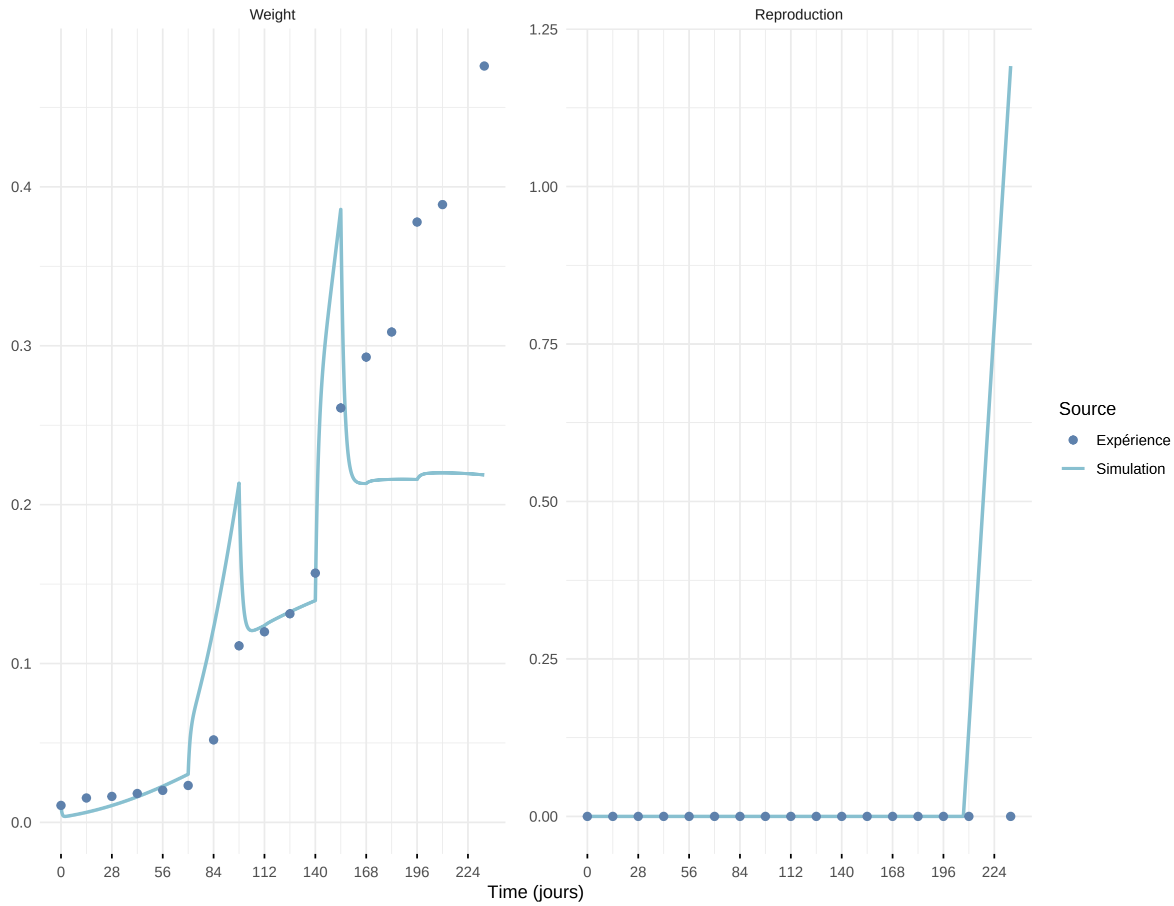
Expérience : Bart2019_XP1_1

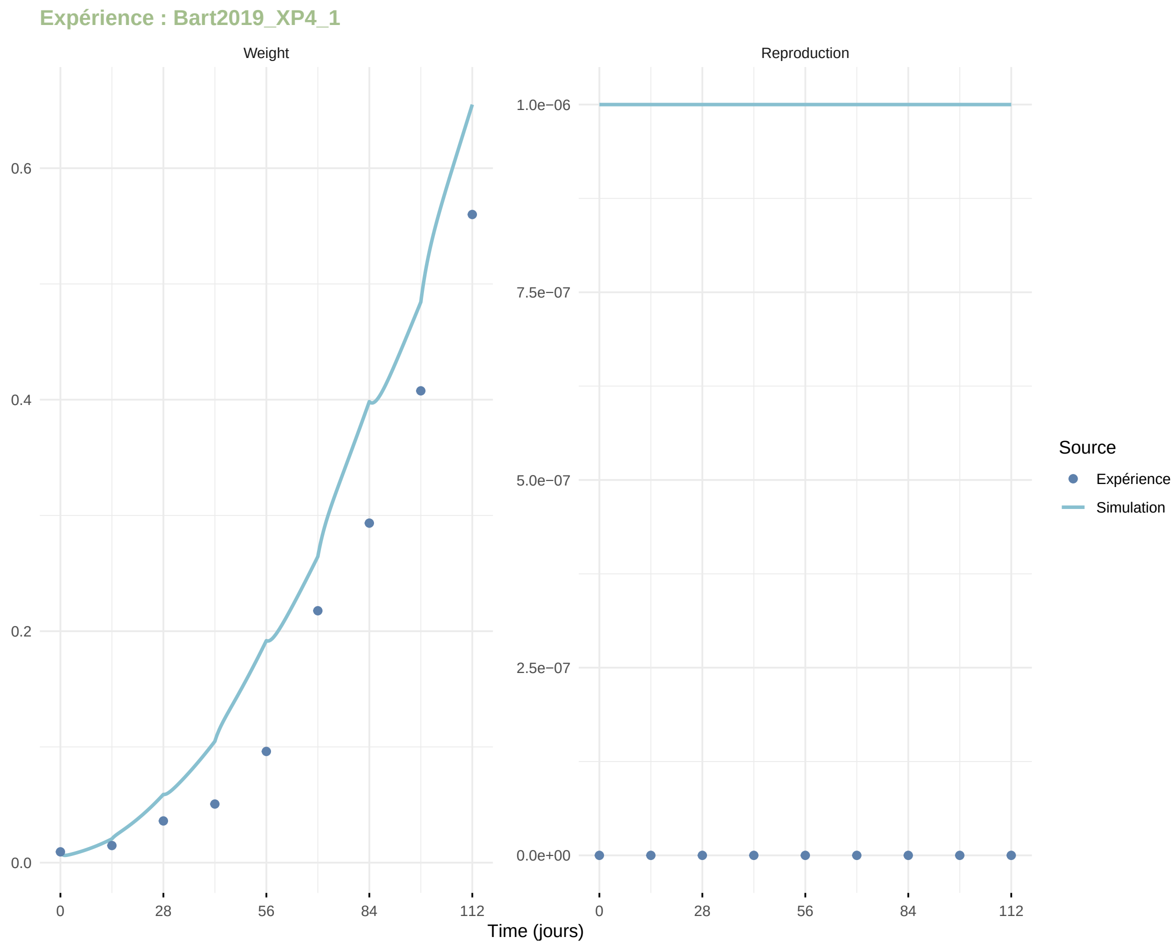


Expérience : Bart2019_XP1_2

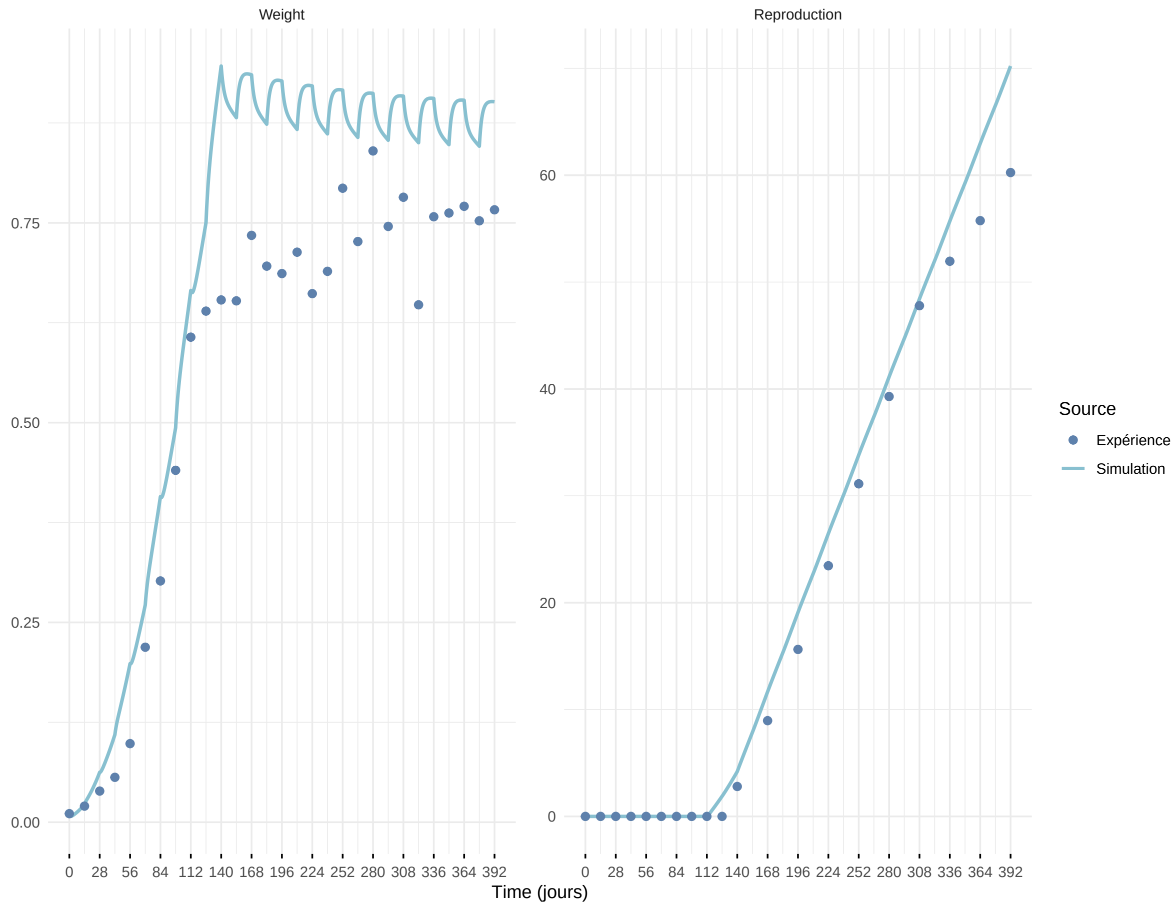


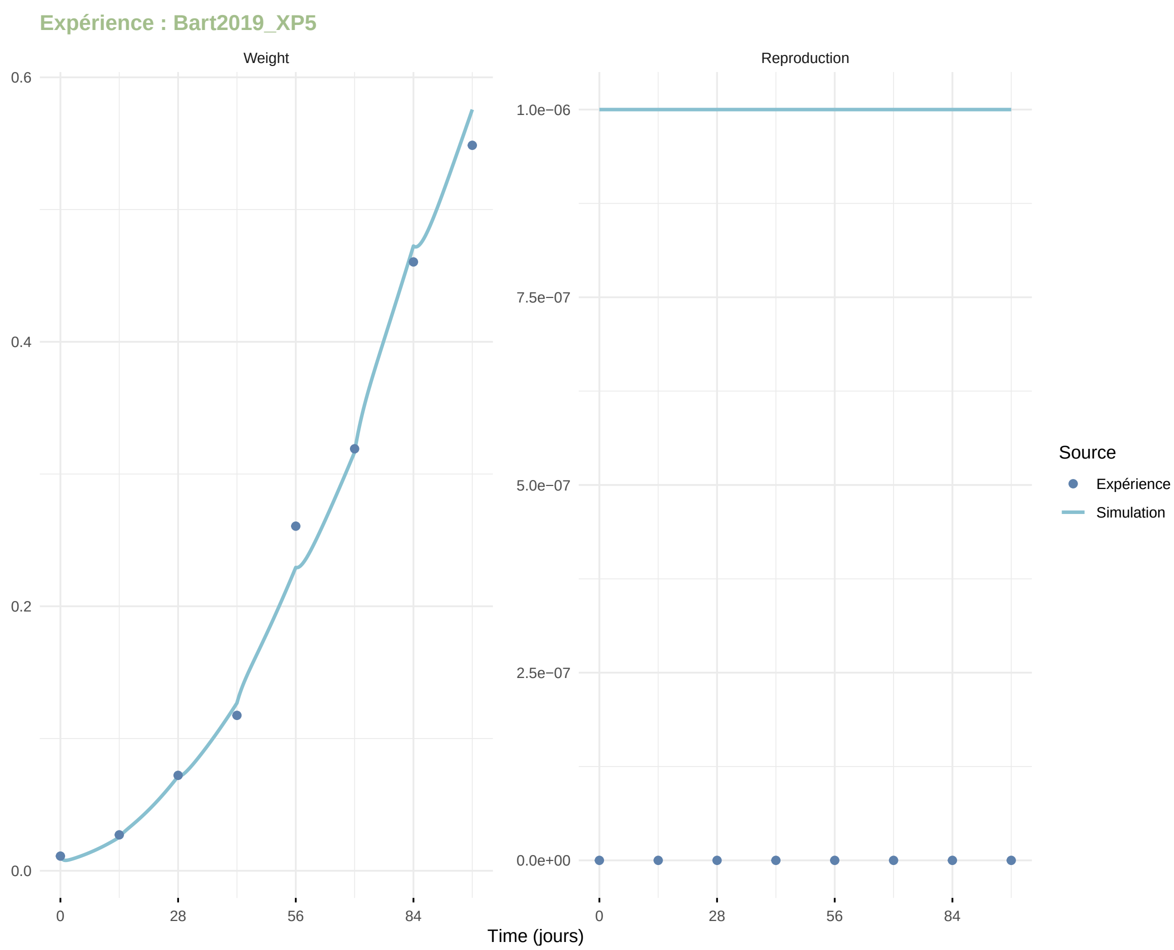
Expérience : Bart2019_XP3

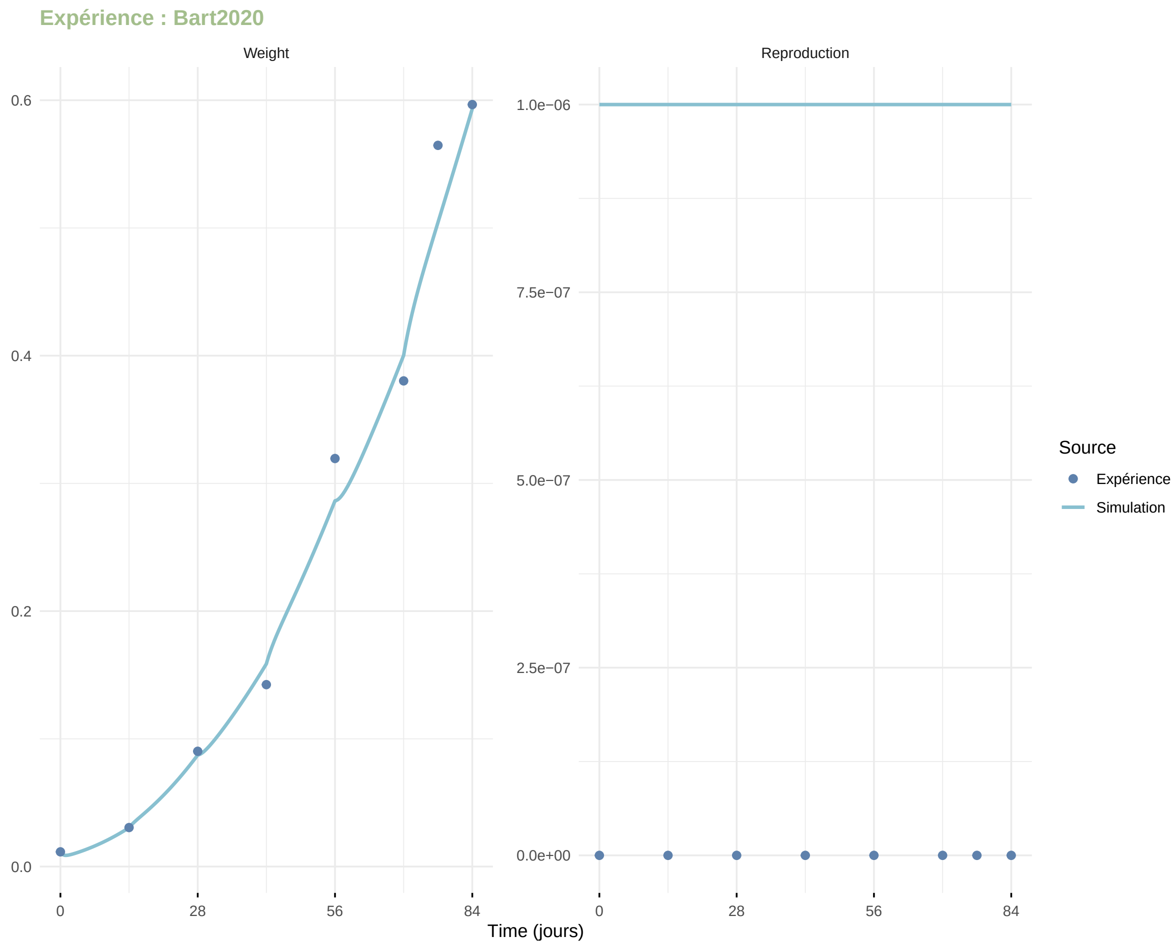


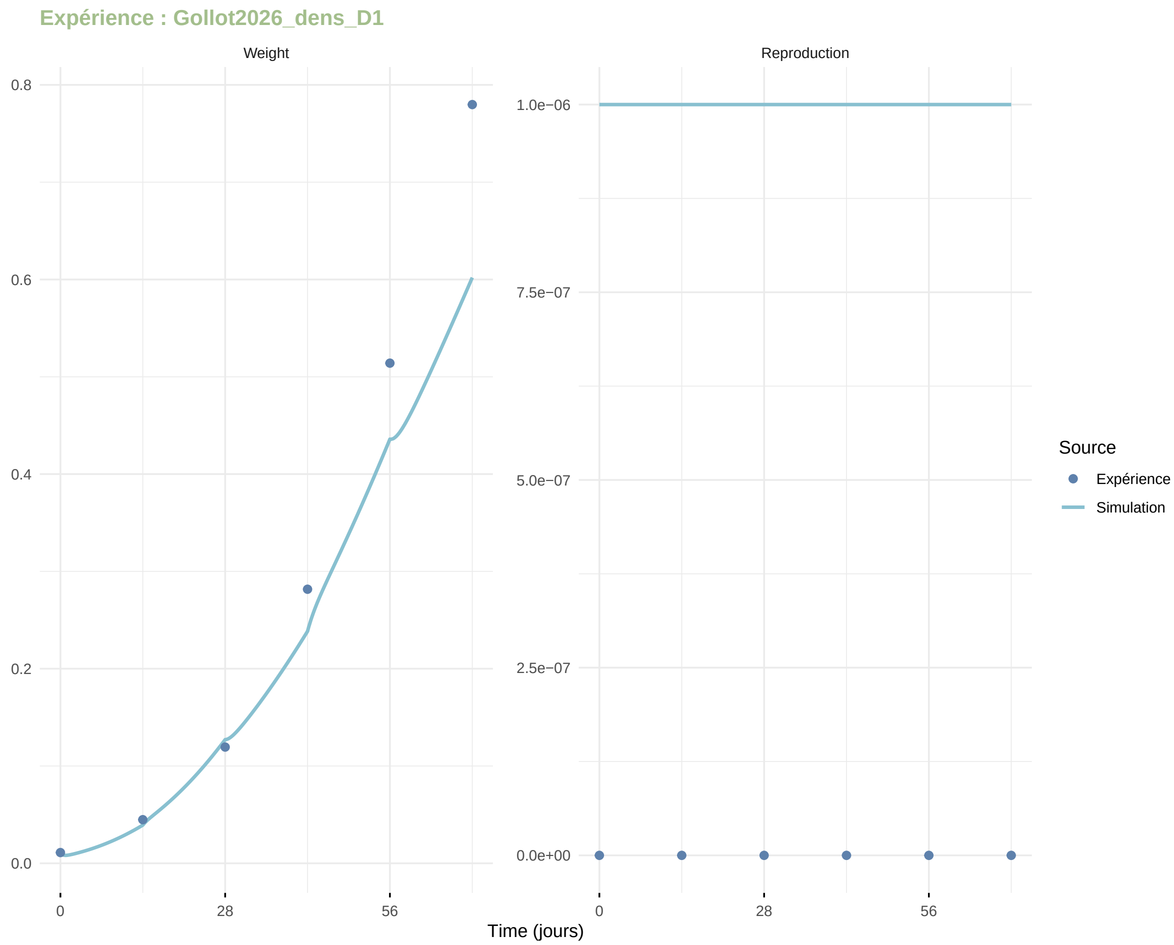

Expérience : Bart2019_XP4_1
Weight
Reproduction
0.0
0.2
0.4
0.6
0.0e+00
2.5e−07
5.0e−07
7.5e−07
1.0e−06
0
28
56
84
112
Time (jours)
Source
Expérience
Simulation

Expérience : Bart2019_XP4_2

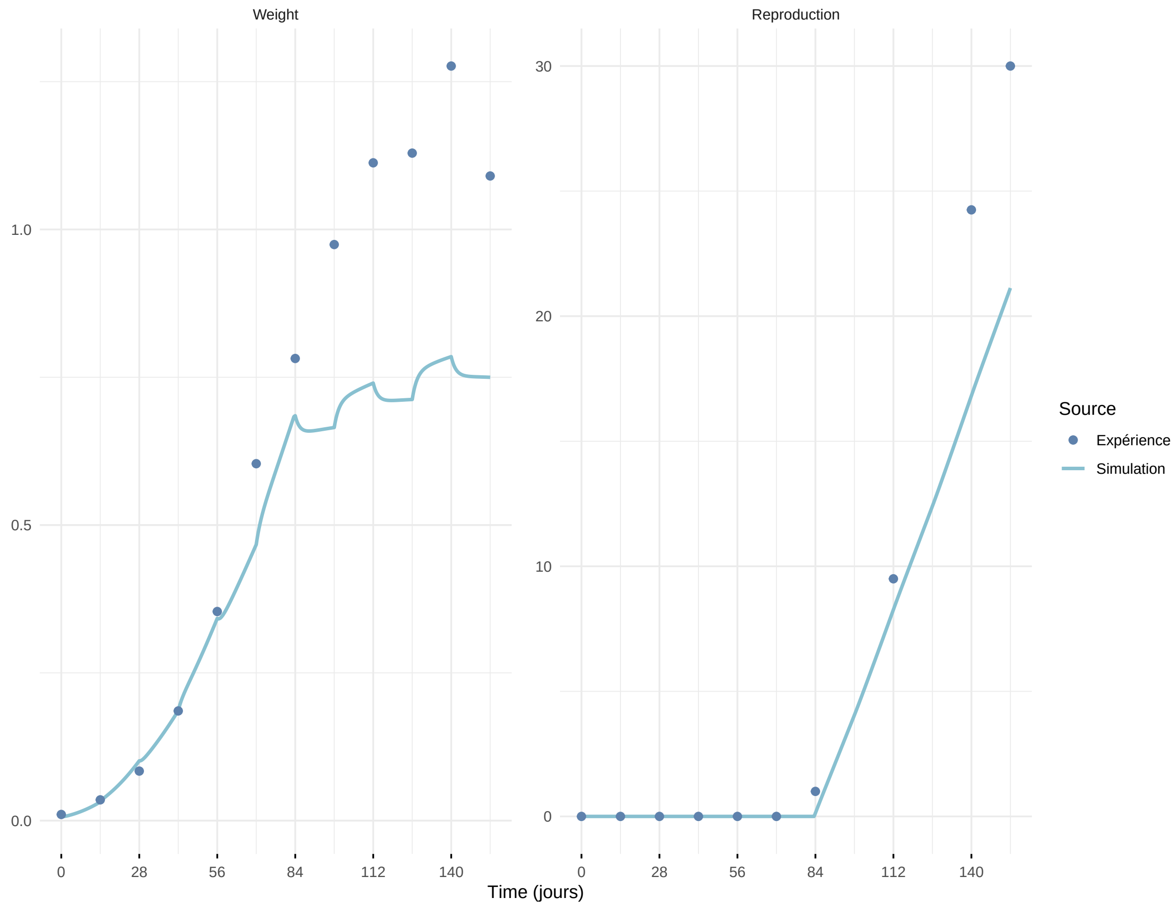


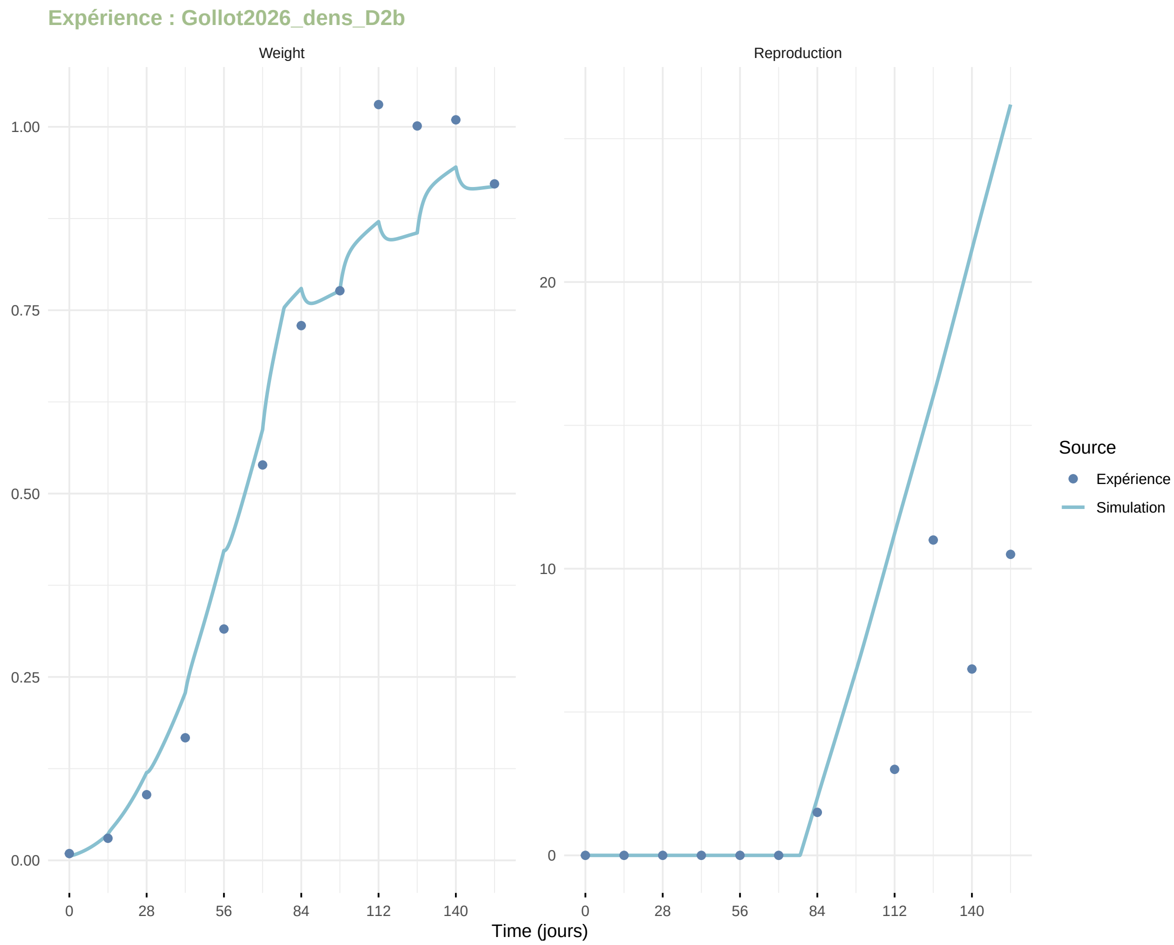

Expérience : Bart2019_XP5
Weight
Reproduction
0.6
0.4
0.2
0.0
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
0
28
56
84
Time (jours)
Source
Expérience
Simulation


Expérience : Bart2020
Weight
Reproduction
0.6
0.4
0.2
0.0
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
0
28
56
84
Time (jours)
Source
Expérience
Simulation

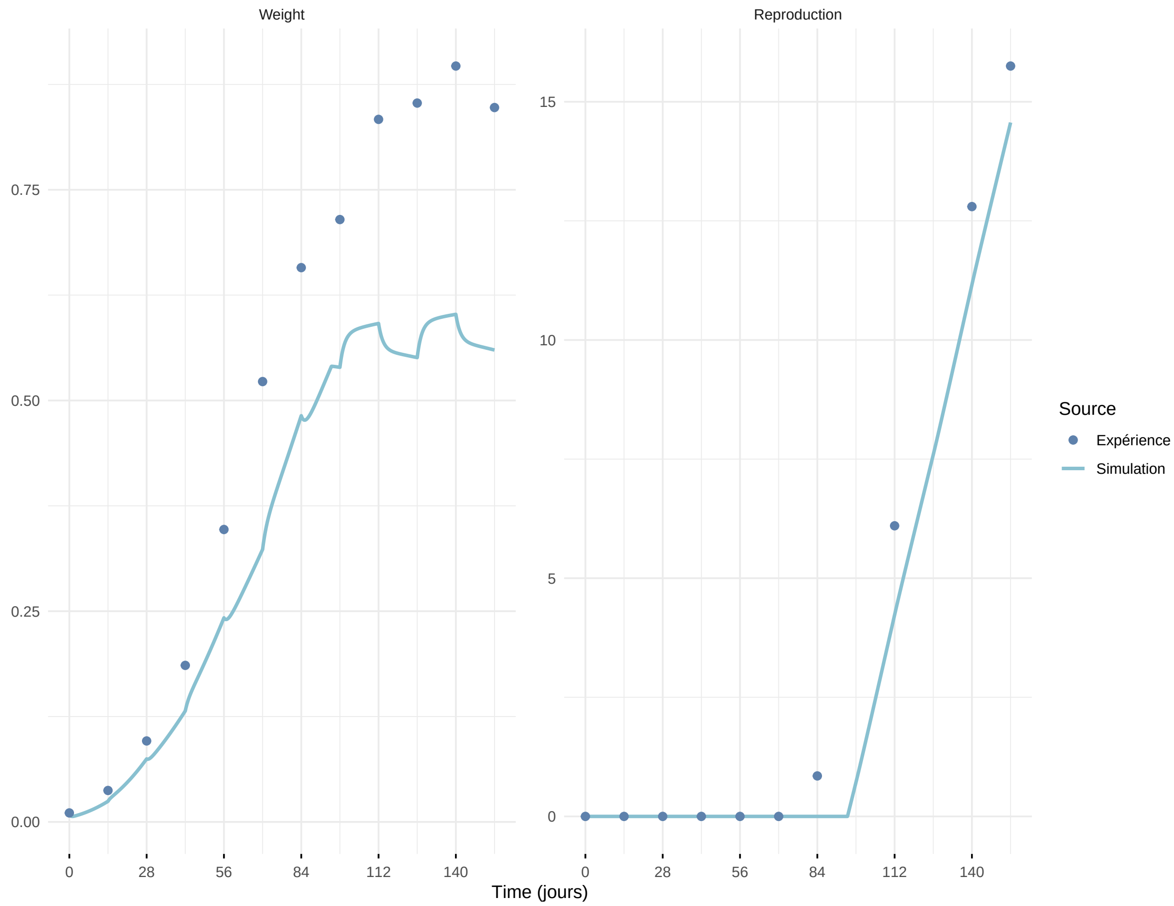

Expérience : Gollot2026_dens_D1
Weight
Reproduction
0.8
0.6
0.4
0.2
0.0
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
0
28
56
Time (jours)
Source
Expérience
Simulation

Expérience : Gollot2026_dens_D2




Expérience : Gollot2026_dens_D2b
Weight
Reproduction
1.00
0.75
0.50
0.25
0.00
20
10
0
0
28
56
84
112
140
Time (jours)
Source
Expérience
Simulation

Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7